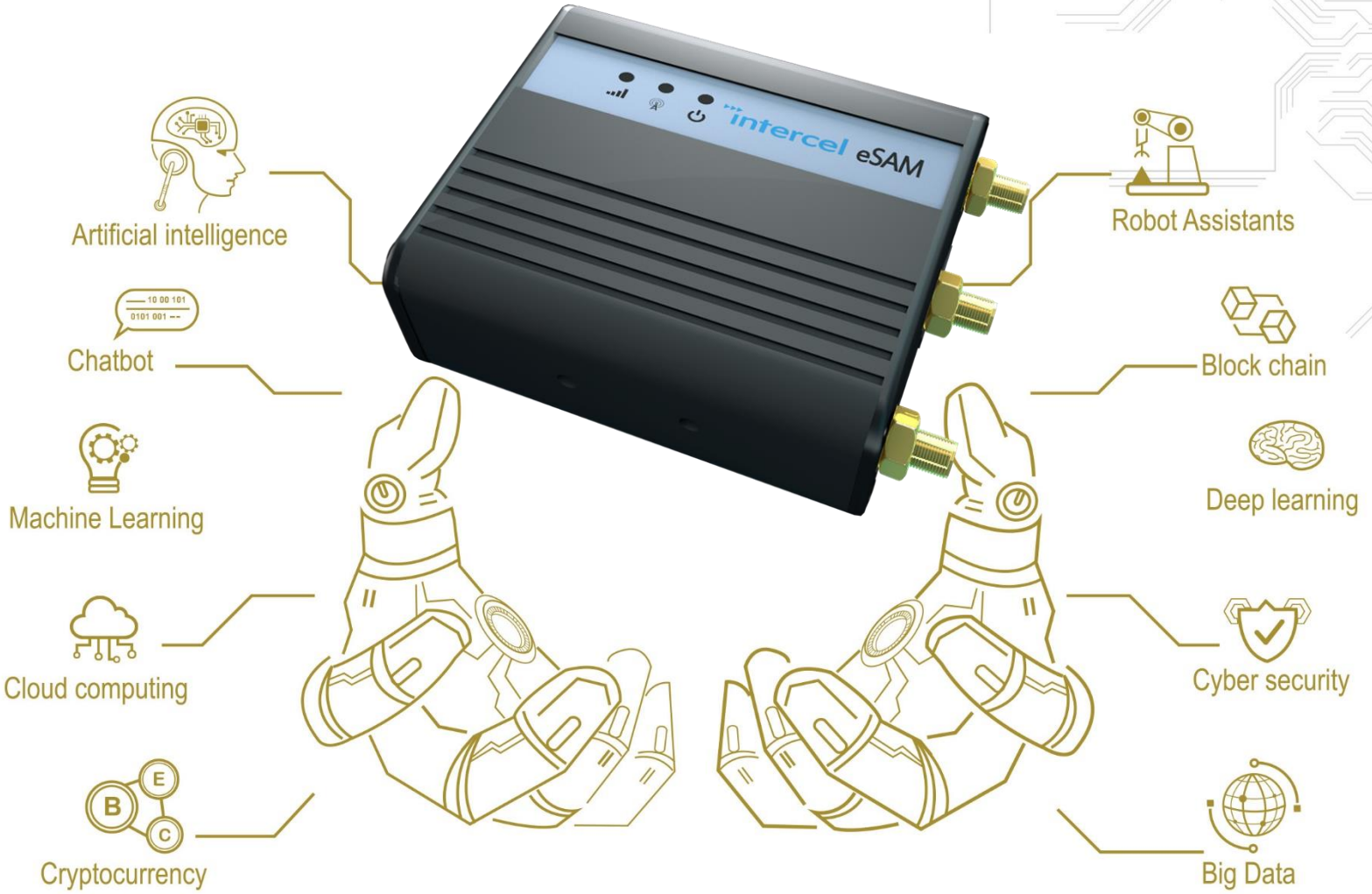
intercel

eSAM & Ultra eSAM

User Guide: MicroBMS Modbus Data Collection



Introduction

Utilising Intercel's MicroBMS Firmware, the eSAM and Ultra eSAM are both capable of collecting Modbus data from a compatible device and saving the data as a .CSV file. Once collection is complete this .CSV file can then be automatically uploaded to an FTP server for the customer.

This guide will explain how to configure Modbus collection settings and send data to the FTP server.

If you have any queries regarding the eSAM, please contact Intercel technical support at intercel@intercel.com.au.

Pre-Requisites

Prior to undertaking the activities outlined in this user guide, please confirm that your eSAM or Ultra eSAM is loaded with the following firmware version:

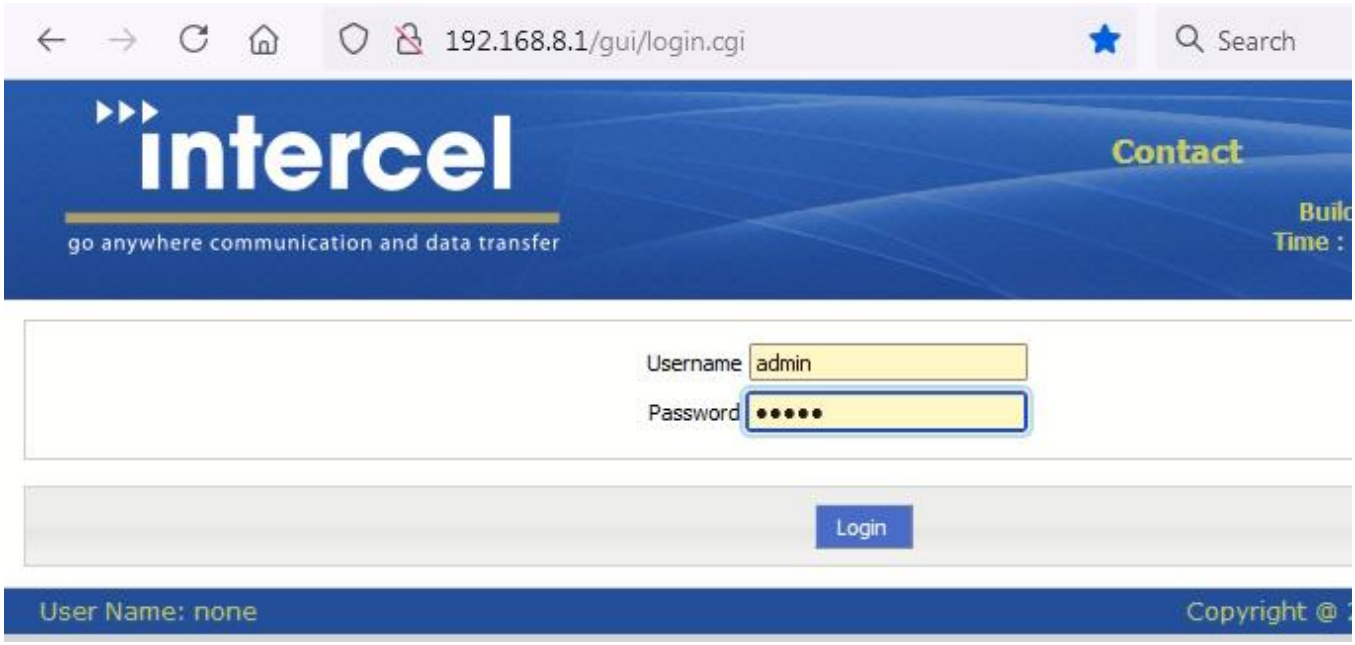
- eSAM_APP_V1.0.5_BMS_2307251403.trx

If your device does not show the same firmware version to that listed above, please contact our support team who can provide the relevant file for upgrading your firmware.

Initial Setup

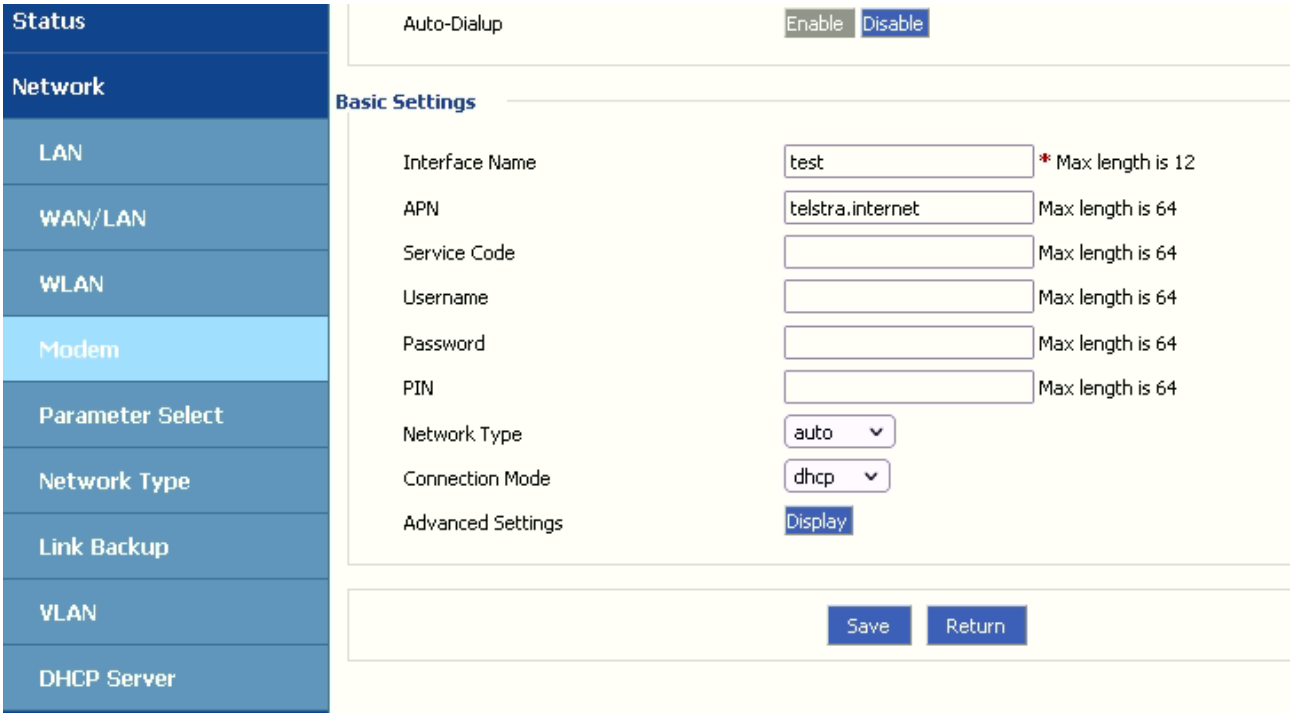
In this section we will configure an eSAM to be easily accessed over the internet.

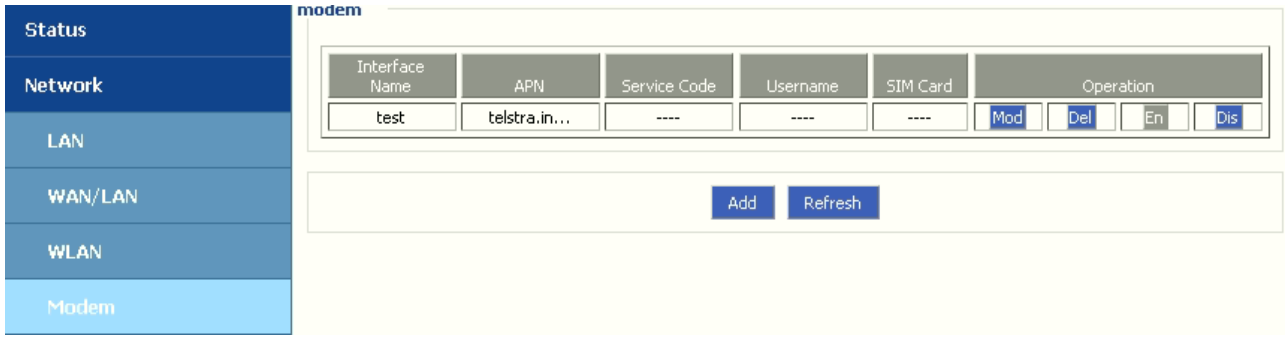
Use your web browser to connect to the router. By default, this can be done by connecting your computer to the eSAM's Ethernet port and entering the IP Address http://192.168.8.1 into your web browser. You will be presented with the screen below.



1. Log into the eSAM / Ultra eSAM. The default login details are:
   Username: admin
   Password: admin
   (You should change this for security reasons.)

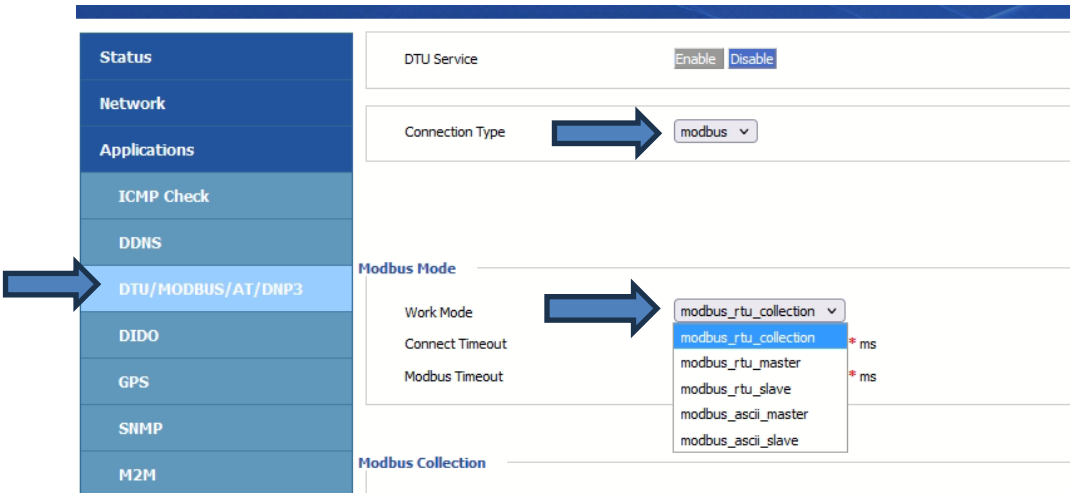
2. Load the APN that you are using for your SIM card's 4G connection.
   Navigate to Network -> Modem.
   Check with you sim card provider for the correct APN for your internet service. Once loaded SAVE the apn. In this example we will load the generic Telstra APN – telstra.internet.


Status
Network
LAN
WAN/LAN
WLAN
Modem
Parameter Select
Network Type
Link Backup
VLAN
DHCP Server
Auto-Dialup
Enable
Disable
Basic Settings
Interface Name
test
* Max length is 12
APN
telstra.internet
Max length is 64
Service Code
Max length is 64
Username
Max length is 64
Password
Max length is 64
PIN
Max length is 64
Network Type
auto
Connection Mode
dhcp
Advanced Settings
Display
Save
Return


Status
Network
LAN
WAN/LAN
WLAN
Modem
modem
Interface Name
APN
Service Code
Username
SIM Card
Operation
test
telstra.in...

Mod
Del
En
Dis
Add
Refresh

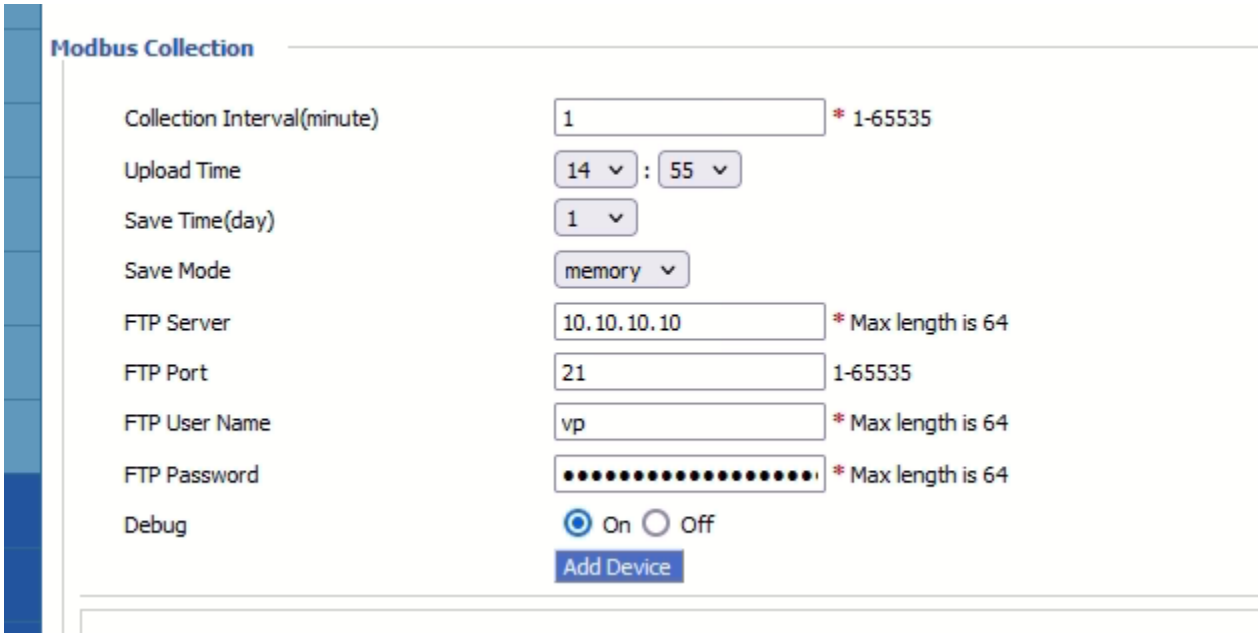
3. Navigate to Applications -> DTU/Modbus/AT/DNP3
   - Connection Type → Modbus
   - Work Mode → Modbus_rtu_collection



4. Load the settings for collection of data.

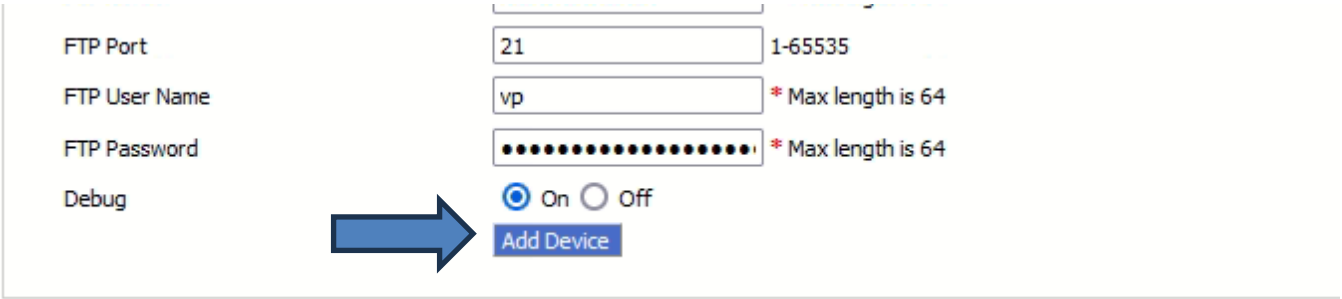
• Collection time interval is when the modem Scans for Modbus data from the serial connection.	In this case we will set the minimum collection time 1 minute.
• Upload Time is the time in 24hr time to upload to the FTP Server.	For testing we will send at 14:55 hours (2:55 pm) to the FTP server.
• Save time.	How long the data will be saved in the router. In this case we will set it to 1 day. Data can be stored for up to 31 days.
• Save mode is the location to save the CSV data before it is uploaded.	For Ultra eSAM we can select USB memory stick or memory. For eSAM we can only select memory. USB Stick must be FAT32 format.
• FTP Server IP or URL address.	Remote FTP Server IP or URL
• FTP PORT.	Port of FTP server. Most FTP ports are 21.
• FTP Username & FTP Password.	Username and Password for access to the FTP server.
• Debug On / Off.	Debug will help us diagnose the logs if needed. As a customer you can leave debugging off.

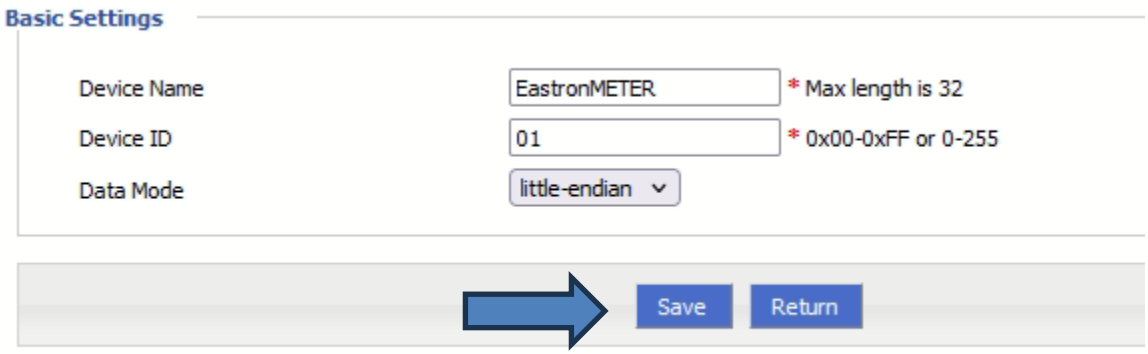
**** Note if uploading fails the upload will retry every minute until the file is uploaded, the modem powers off or the feature is disabled.**

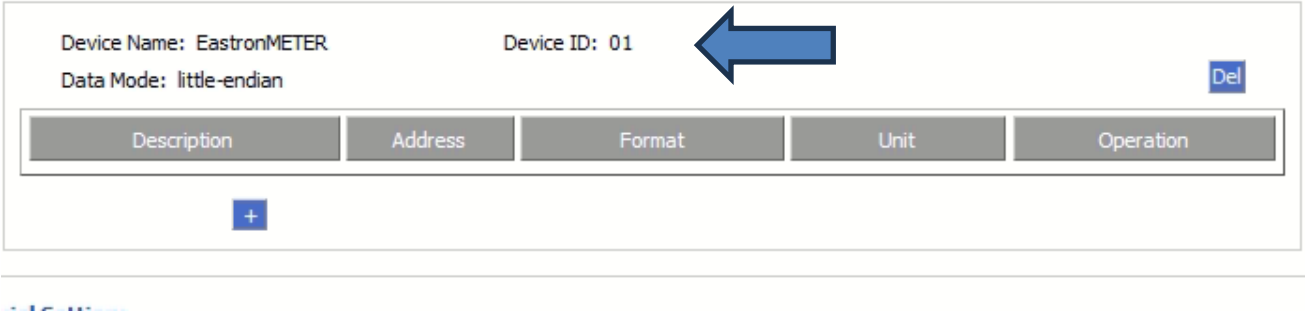


5. Now we will add the device. For this test we are connecting to an Eastron Meter.

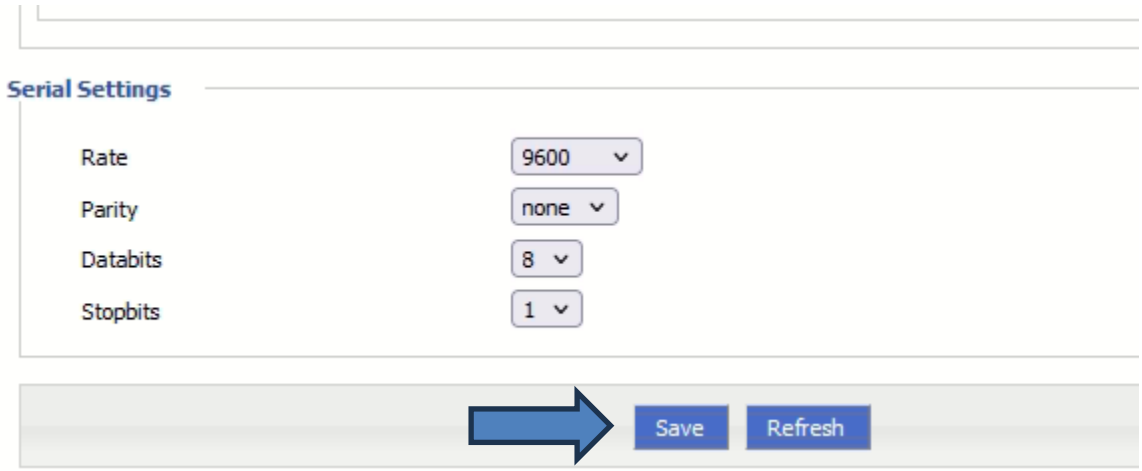
- The device name is a free text field. In this test we will use the manufacturer Eastron METER.
- The device ID needs to be correct according to the SN or the SN programmed in the meter. In this case the SN is 01 for this Meter. If connecting devices in a daisy chain the device IDs must be unique to each device in the chain.
- Device ID range = 0-255 for Decimal. 0x00 – 0xFF for Hexadecimal.
- Data mode is Little-endian or Big-endian. Check the manual for your Meter. In this meter we are using Little-endian. Once entered, save this device.



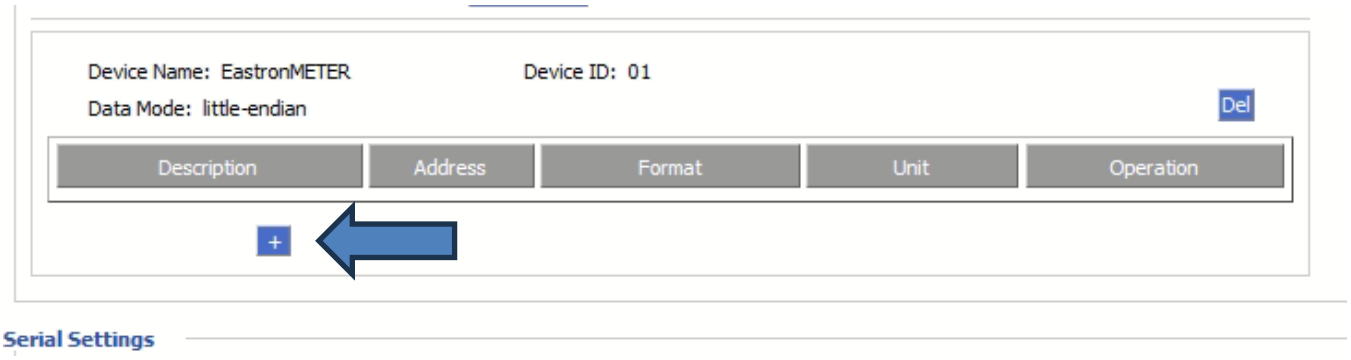




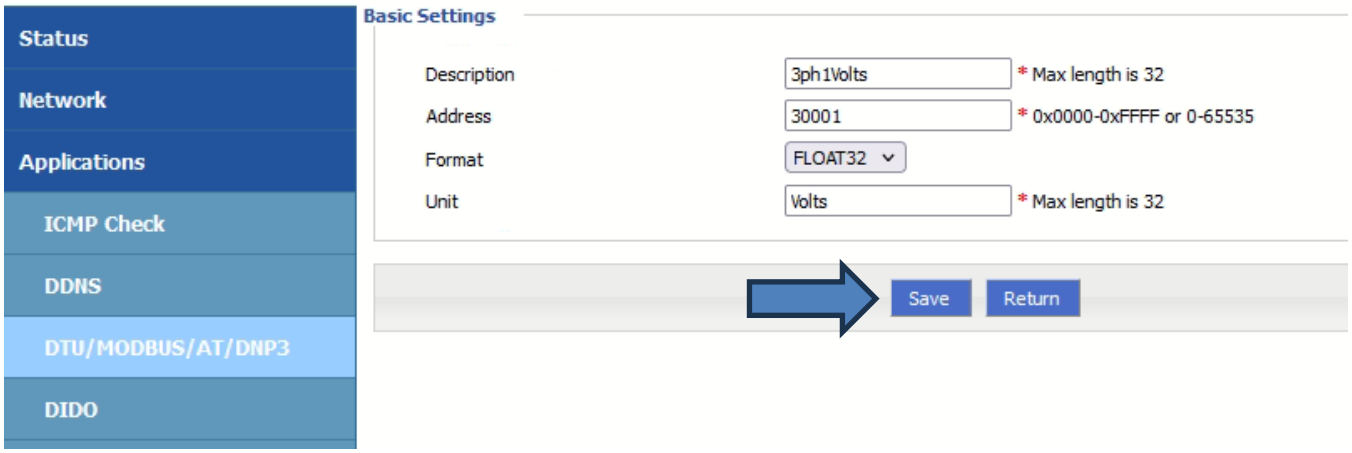
6. Prior to entering the Modbus registers we need to set the serial speed and data settings first.
   - For this device we are using standard serial settings. Save once the settings are set. 9600/None/8/1



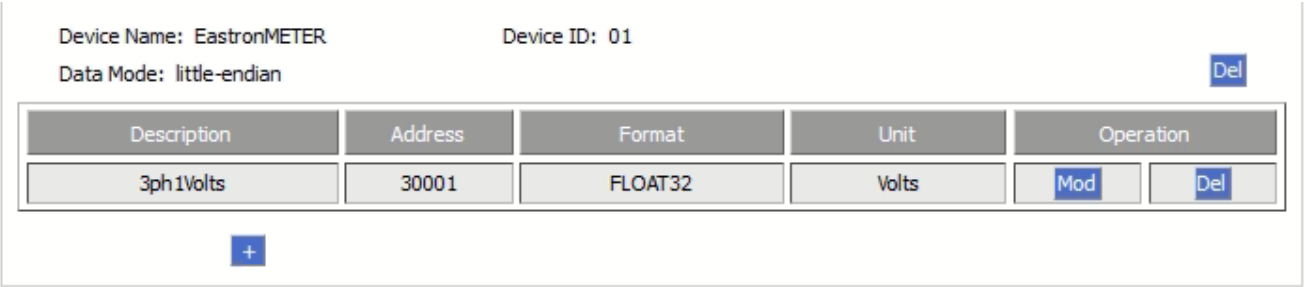
7. For testing we need to load the Modbus registers that wish to collect data from. We will load as many as possible for testing and proof of function. The meter has 240VAC connected to 3 of the phases.
   - Under the loaded device click the + icon to load each register.



8. We can load the Registers we want to read. Refer to the Modbus register map of the device you wish to collect data from. The first register will be Volts on Phase 1 of the Meter.
   - Description names the measurement.
   - Modbus address refers to the register address. 0x0000-0xFFFF is Hexadecimal. 0-65535 is Decimal.
   - Format for the Data Format. Float, BIN, HEX, UINT (unsigned Integer), INT and Variations for 16 & 32 Bit.
   - Unit name for value. For this first Register we are measuring Volts.
   - Save one completed.



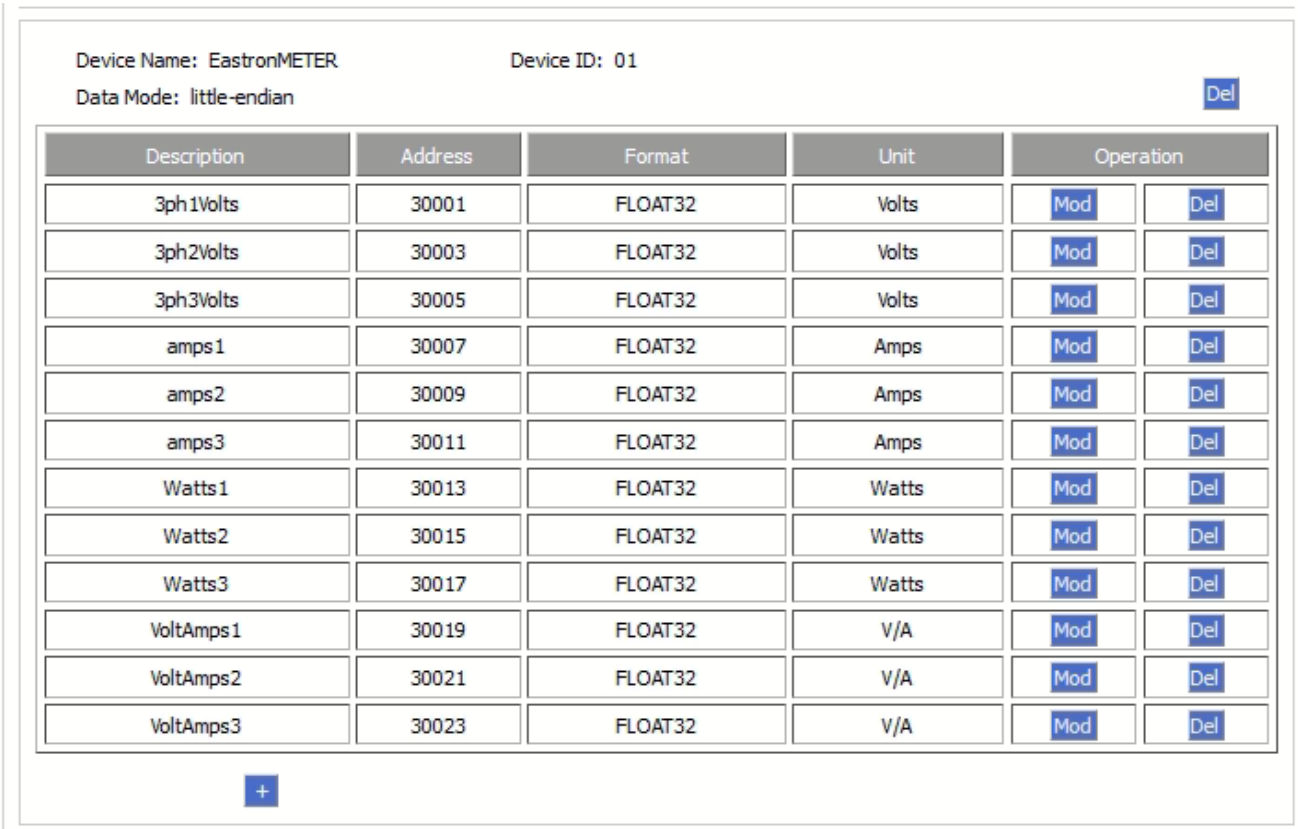
9. Once saved the new address will be saved under the device, shown below.



10. Repeat the previous steps to add more registers as required. For our testing, we have configured registers for:

    Volts, Amps, Watts, and Volt-Amperes for all 3 Phases. Our test hardware is set up with voltage only on the first 2 phases, therefore we expect some of the fields will show 0 values.

-

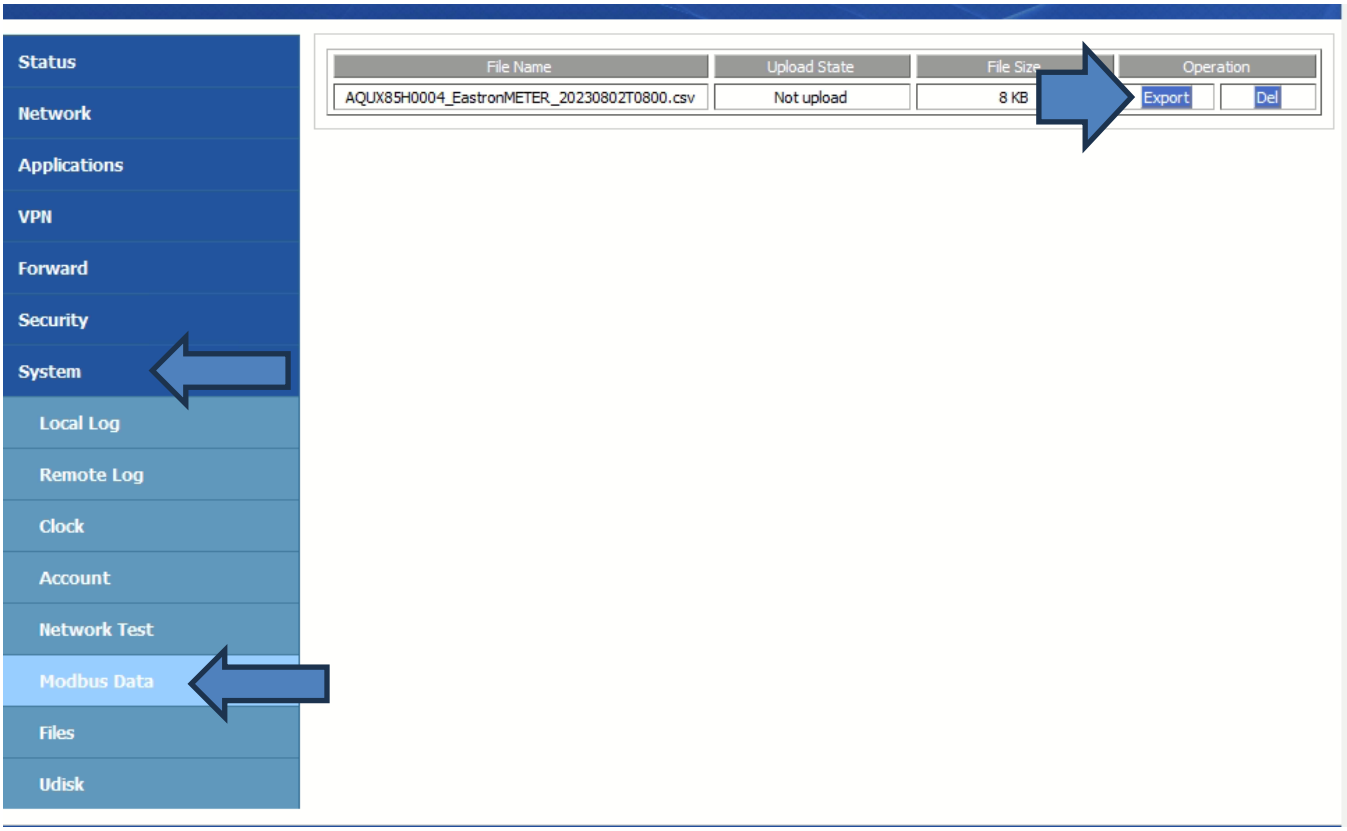


The following extract shows a portion of the Modbus register map for our test device, the Address fields correspond with those which we have configured in the eSAM.

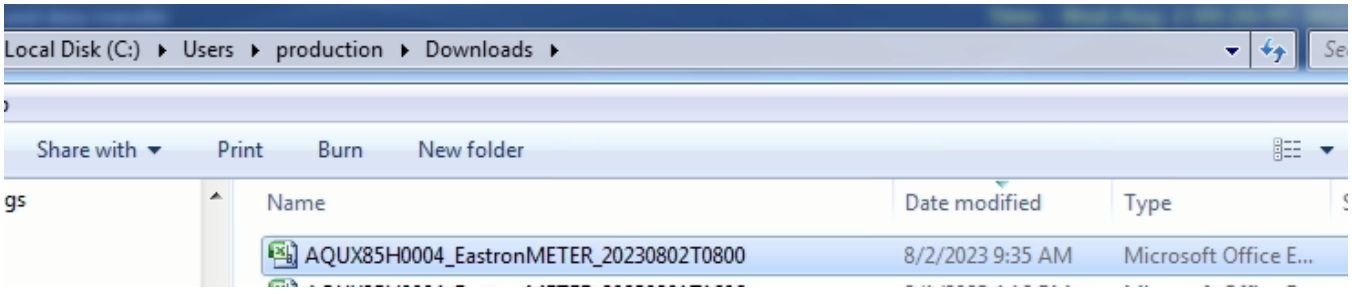
Address (Register)	Parameter Number	SDM630Modbus Input Register Parameter		Modbus Protocol Start Address Hex		3 Ø	3 Ø	1 Ø
		Description	Units	Hi Byte	Lo Byte	4 W	3 W	2 W
30001	1	Phase 1 line to neutral volts.	Volts	00	00	√	X	√
30003	2	Phase 2 line to neutral volts.	Volts	00	02	√	X	X
30005	3	Phase 3 line to neutral volts.	Volts	00	04	√	X	X
30007	4	Phase 1 current.	Amps	00	06	√	√	√
30009	5	Phase 2 current.	Amps	00	08	√	√	X
30011	6	Phase 3 current.	Amps	00	0A	√	√	X
30013	7	Phase 1 power.	Watts	00	0C	√	X	√
30015	8	Phase 2 power.	Watts	00	0E	√	X	√
30017	9	Phase 3 power.	Watts	00	10	√	X	X
30019	10	Phase 1 volt amps.	VA	00	12	√	X	√
30021	11	Phase 2 volt amps.	VA	00	14	√	X	X
30023	12	Phase 3 volt amps.	VA	00	16	√	X	X

Testing

11. Once the meter is connected for a given period, we can Export the .CSV file before its uploaded to the FTP server to confirm the data is correct.
    - Navigate to System→Modbus Data. The file will be ready to view. You can Export the file to examine the data.



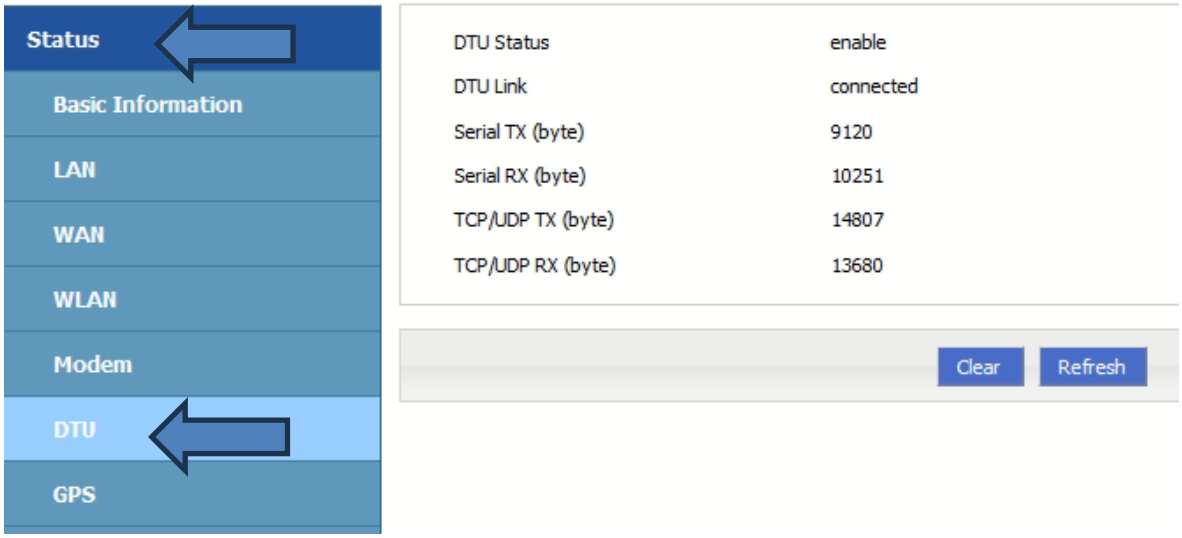
12. The File is exported to the local hard drive for viewing.



13. We can open the .CSV file in Excel and adjust the columns accordingly for viewing.
    - We need to note that no load is connected to the meter. Also remember that no voltage is connected to Phase 3 of the meter. Only a small amount of idle current is measurable for the phase 2 circuit in the meter. A real-world configuration would look different to this extract.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Description	3ph1Volts	3ph2Volts	3ph3Vol	amps1	amps2	amps3	Watts1	Watts2	Watts3	VoltAmp:	VoltAmps2	VoltAmps3
2	Address	30001	30003	30005	30007	30009	30011	30013	30015	30017	30019	30021	30023
3	Data Format	FLOAT32	FLOAT32	FLOAT32	FLOAT3	FLOAT32	FLOAT	FLOAT3	FLOAT3	FLOAT3	FLOAT32	FLOAT32	FLOAT32
4	Data Unit	Volts	Volts	Volts	Amps	Amps	Amps	Watts	Watts	Watts	V/A	V/A	V/A
5	1/8/2023 7:59	232.679	232.7263	0	0	0.07031	0	0	0	0	0	15.375	0
6	2/8/2023 8:01	232.2731	232.4895	0	0	0.07031	0	0	0	0	0	15.375	0
7	2/8/2023 8:02	231.5009	231.7242	0	0	0.07031	0	0	0	0	0	15.3125	0
8	2/8/2023 8:03	232.5387	232.395	0	0	0.07031	0	0	0	0	0	15.1875	0
9	2/8/2023 8:05	232.0336	231.1978	0	0	0.07031	0	0	0	0	0	15.3125	0
10	2/8/2023 8:06	231.5825	231.1002	0	0	0.07031	0	0	0	0	0	15.3125	0
11	2/8/2023 8:08	231.6527	231.5208	0	0	0.07031	0	0	0	0	0	15.3125	0
12	2/8/2023 8:09	231.4724	231.481	0	0	0.07031	0	0	0	0	0	15.3125	0
13	2/8/2023 8:10	231.4299	231.4236	0	0	0.07031	0	0	0	0	0	15.25	0
14	2/8/2023 8:12	230.8596	230.0806	0	0	0.07031	0	0	0	0	0	15.3125	0
15	2/8/2023 8:13	230.9626	230.9736	0	0	0.0708	0	0	0	0	0	15.3125	0
16	2/8/2023 8:14	231.2417	230.1479	0	0	0.0708	0	0	0	0	0	15.25	0
17	2/8/2023 8:16	231.6316	231.6837	0	0	0.0708	0	0	0	0	0	15.4375	0
18	2/8/2023 8:17	231.787	231.9512	0	0	0.07129	0	0	0	0	0	15.375	0
19	2/8/2023 8:19	230.9435	230.7816	0	0	0.0708	0	0	0	0	0	15.375	0
20	2/8/2023 8:20	231.6904	231.6919	0	0	0.0708	0	0	0	0	0	15.4375	0
21	2/8/2023 8:21	231.8247	231.8525	0	0	0.07129	0	0	0	0	0	15.5	0
22	2/8/2023 8:23	231.3564	230.8886	0	0	0.0708	0	0	0	0	0	15.375	0
23	2/8/2023 8:24	231.6233	231.5419	0	0	0.07129	0	0	0	0	0	15.5625	

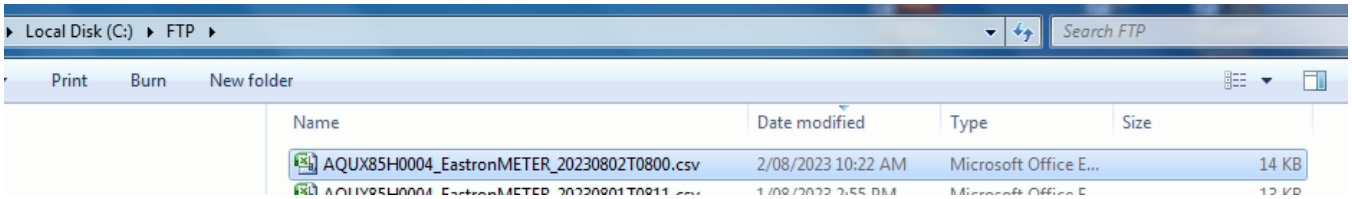
14. We can navigate to Status -> DTU to show data, is being transmitted and received on the serial port.



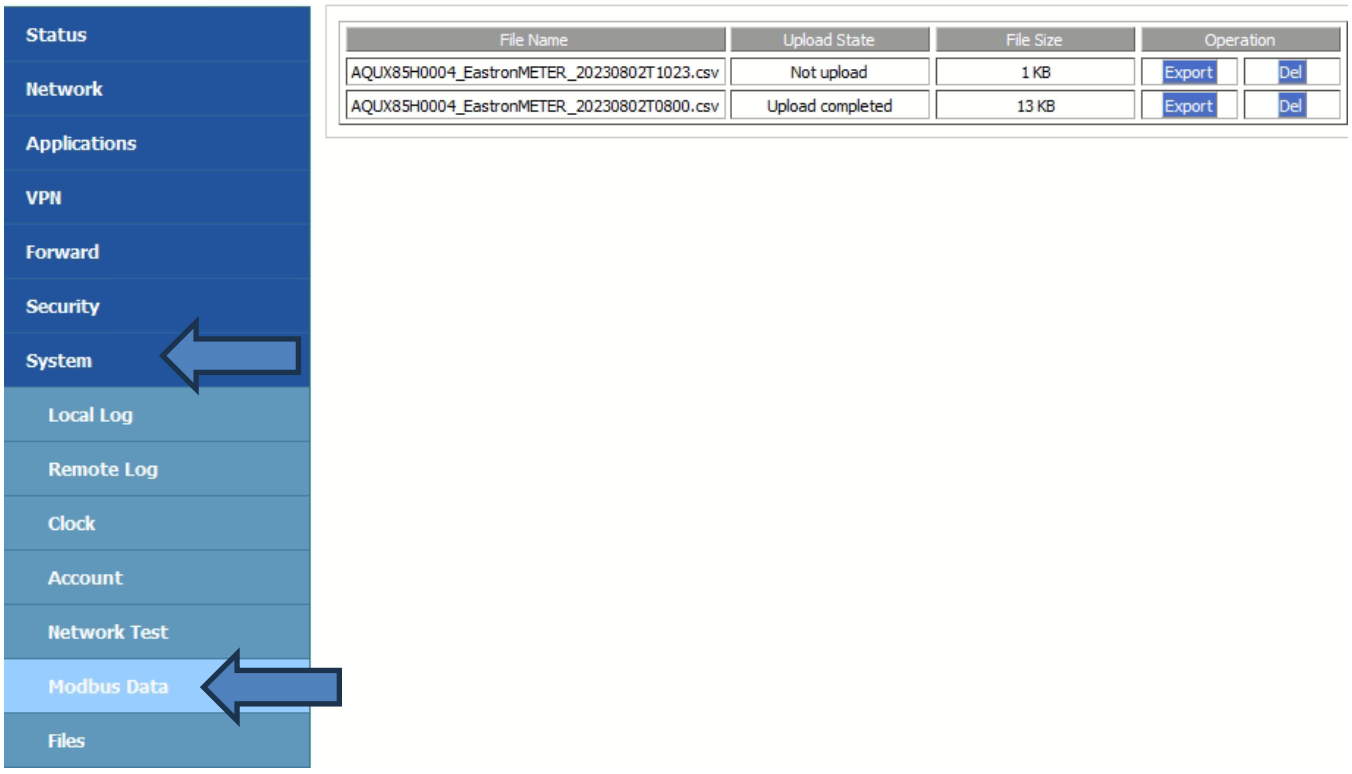
15. Let us check that the .CSV file is uploaded to our test FTP server.
    - We can check the local logs: System -> Local Log -> Message.

```
router daemon.info modbus_collection[4877]: Send >>> (00 21 00 00 00 06 01 04 00 11 00 02 ){modbus_func.c->25}
router daemon.info modbus_collection[4877]: FTP upload timeup(10:22){modbus_ftp.c->67}
router daemon.info modbus_collection[4877]: ftp cmd: ftpput -u vp -p test -P 21 [redacted].176.102 AQUX85H0004_Eastr
```

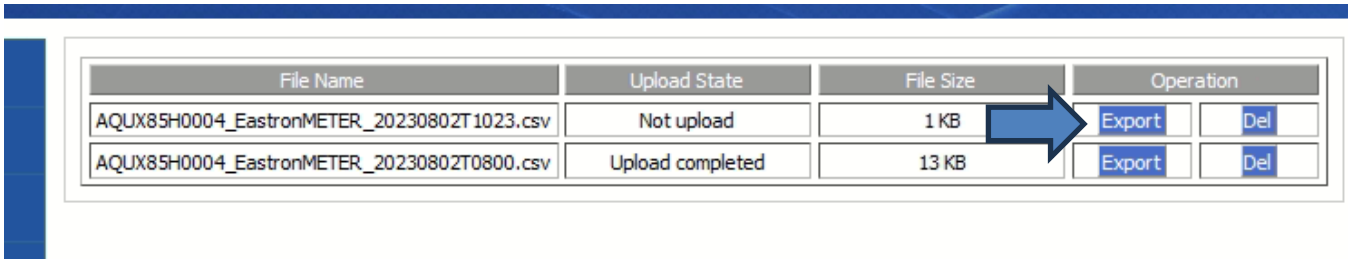
    - We can check the FTP server to confirm the file is uploaded correctly.



16. Let us check the status of the .CSV files now that the upload is complete.
    - Navigate to System -> Modbus Data and confirm 1 file uploaded and 1 new file collecting data. Each file has the eSAM serial number, device name (user configured in previous steps) and is time stamped with a date and 24-hour time of when the .CSV file was created.
        - 2023 year, 08 month, 02 day, T0800 time 08:00 am – File uploaded as the first file we created.
        - 2023 year, 08 month, 02 day, T1023 time 10:23 am – File not uploaded as still collecting data.



17. We can open the new .CSV file collecting data to confirm the time in the file is new.
    - Export file ending T1023.CSV



File Name	Upload State	File Size	Operation	
AQUX85H0004_EastronMETER_20230802T1023.csv	Not upload	1 KB	Export	Del
AQUX85H0004_EastronMETER_20230802T0800.csv	Upload completed	13 KB	Export	Del

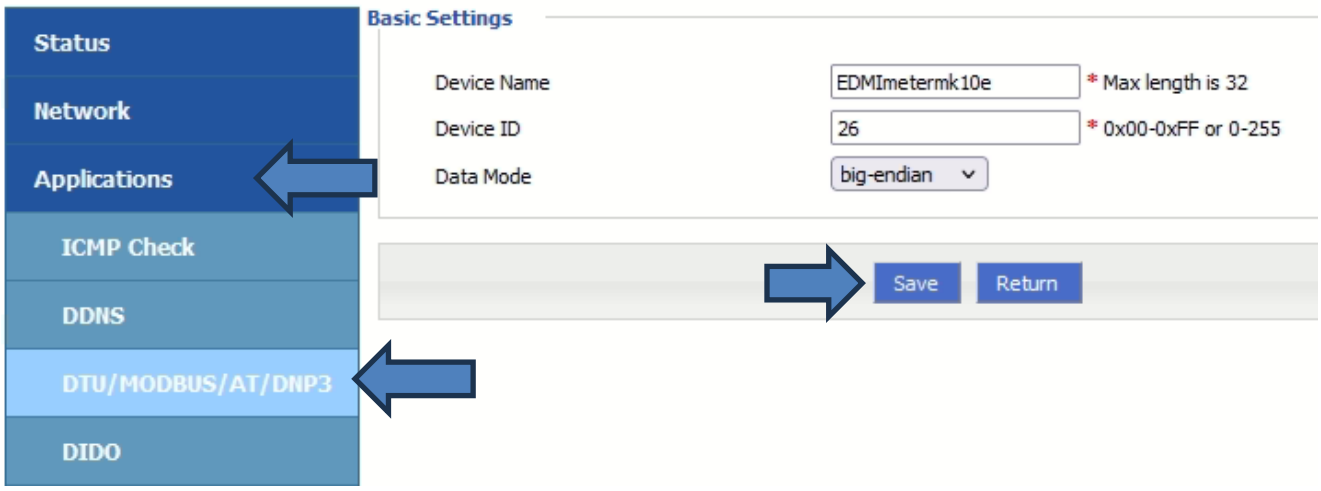
    - The file confirms the new time accordingly 10:22am

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Description	3ph1Volt	3ph2Vol	3ph3V	amps1	amps2	amps3	Watts1	Watts2	Watts3	VoltAmp	VoltAm	VoltAmps3
2	Address	30001	30003	30005	30007	30009	30011	30013	30015	30017	30019	30021	30023
3	Data Format	FLOAT32	FLOAT32	FLOAT	FLOAT3	FLOAT3	FLOAT3	FLOAT3	FLOAT	FLOAT3	FLOAT32	FLOAT3	FLOAT32
4	Data Unit	Volts	Volts	Volts	Amps	Amps	Amps	Watts	Watts	Watts	V/A	V/A	V/A
5	2/8/2023 10:22	228.567	228.547	0	0	0.072	0	0	0	0	0	15.563	0
6	2/8/2023 10:24	228.106	228.332	0	0	0.072	0	0	0	0	0	15.438	0
7	2/8/2023 10:25	228.649	228.808	0	0	0.072	0	0	0	0	0	15.438	0
8	2/8/2023 10:26	226.08	225.027	0	0	0.071	0	0	0	0	0	15.188	0
9	2/8/2023 10:28	226.255	226.251	0	0	0.071	0	0	0	0	0	15.188	0
10	2/8/2023 10:29	226.54	226.561	0	0	0.071	0	0	0	0	0	15.188	0
11	2/8/2023 10:30	226.595	226.634	0	0	0.071	0	0	0	0	0	15.25	0
12	2/8/2023 10:32	226.528	226.568	0	0	0.071	0	0	0	0	0	15.125	0
13	2/8/2023 10:33	226.053	225.871	0	0	0.071	0	0	0	0	0	15.5	0
14	2/8/2023 10:35	228.783	228.911	0	0	0.072	0	0	0	0	0	15.625	0
15	2/8/2023 10:36	232.232	232.384	0	0	0.073	0	0	0	0	0	16	0
16	2/8/2023 10:37	232.232	231.082	0	0	0.073	0	0	0	0	0	16.125	0

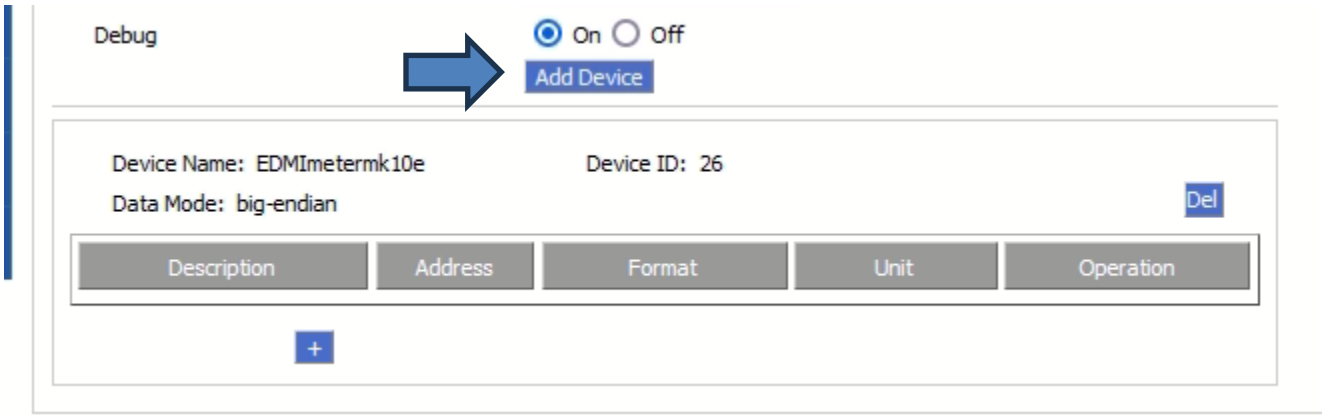
Testing RS485 – Daisy Chain 2 Devices

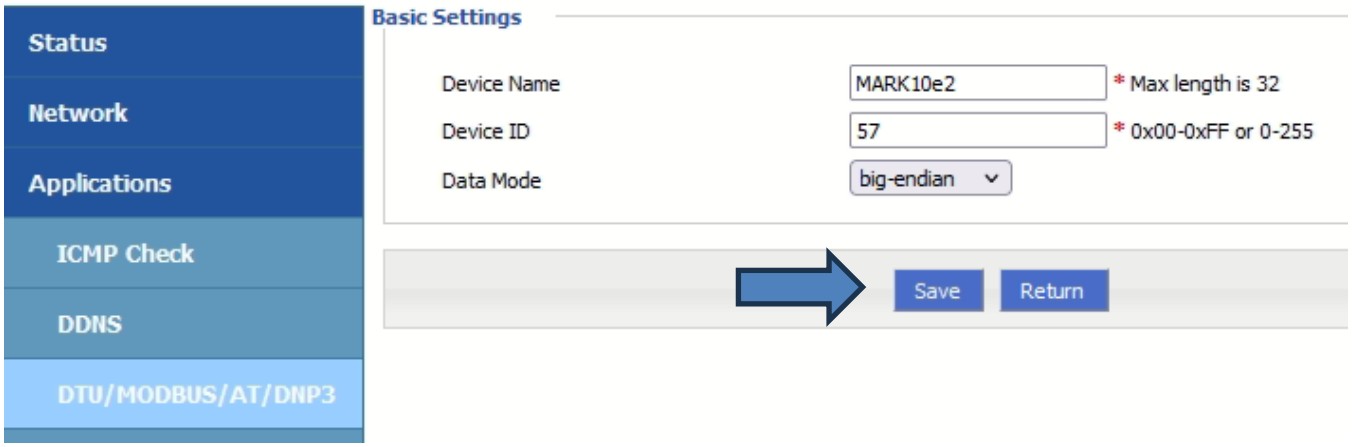
Let us test the feature further for proof of functionality. We will connect 2 meters with RS485 daisy chain.

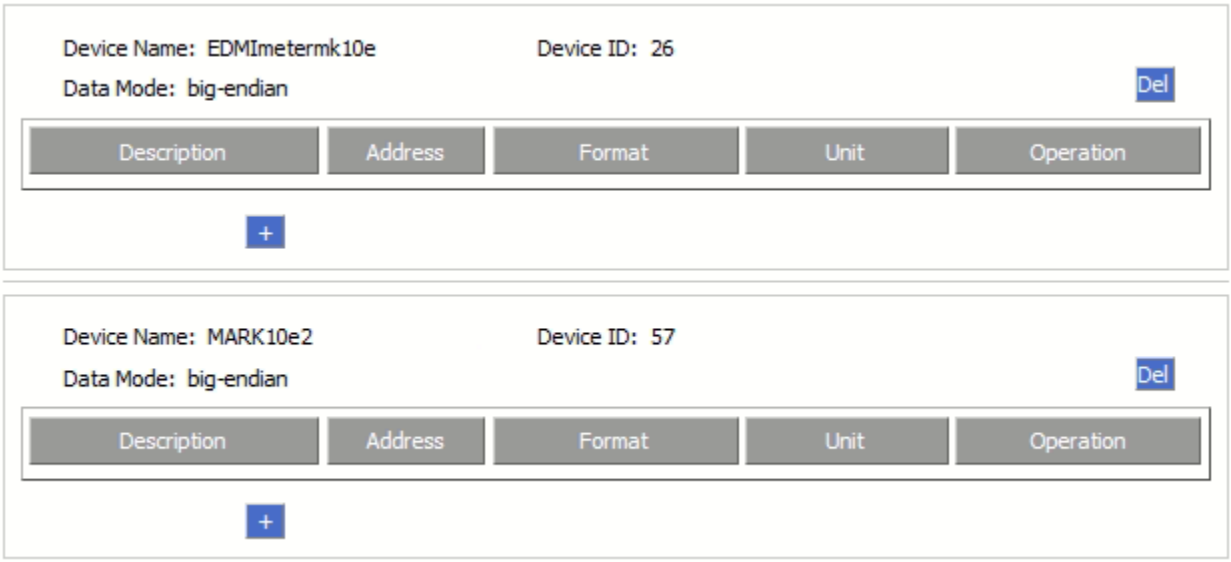
1. Navigate to Applications -> DTU -> add device. Delete the previous device if you are creating a new connection.
   - Device Name: EDMImetermk10e
   - Device ID: 26 (the last 2 numbers of the SN)
   - Data Mode: Big-endian (check the manual for the device).
   - SAVE once completed.



2. We can add the next device to show 2 separate devices in the GUI.
   - Add device number 2
   - Device Name= MARK10e2
   - Device ID= 57 (last 2 numbers of the SN)
   - Data Mode= big-endian



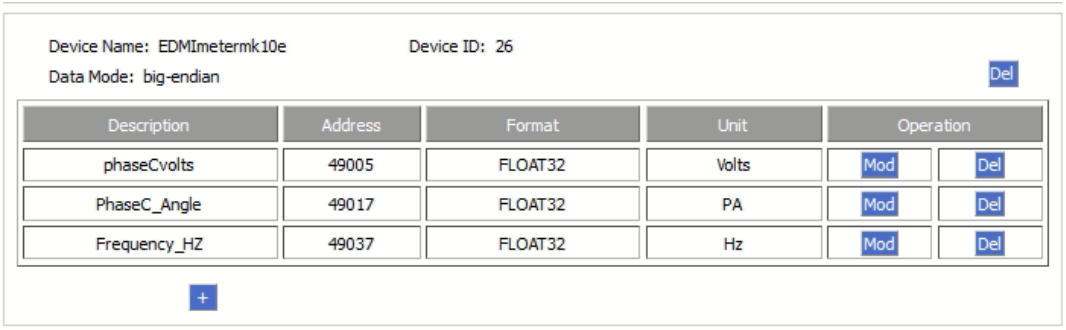


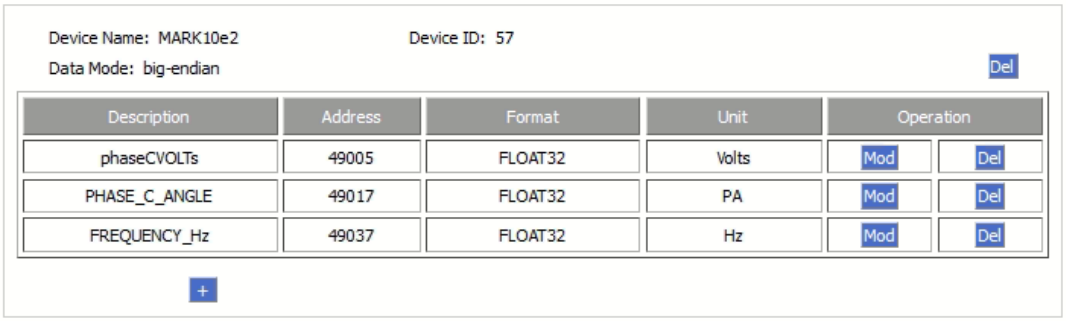


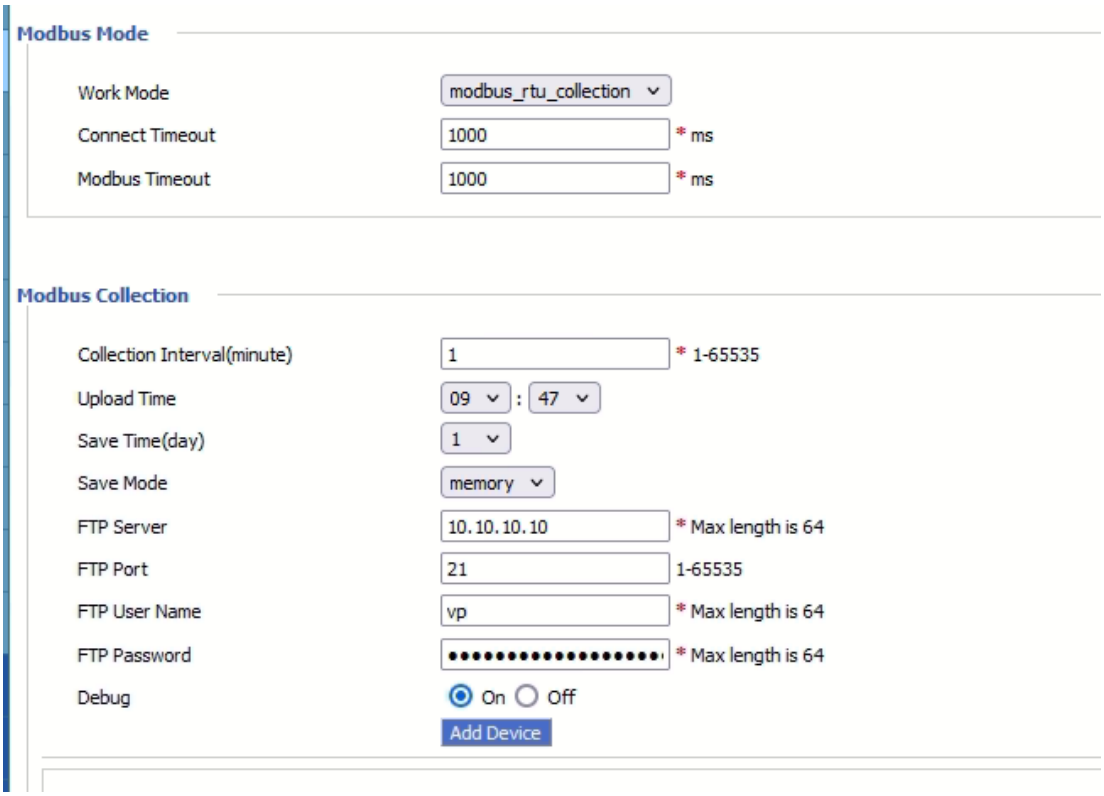
3. Now with 2 separate devices loaded (both the same model) we can assign the data types and registers for each meter. Click the + icon directly under the device loaded.
   - Let us add the registers for device ID 26.



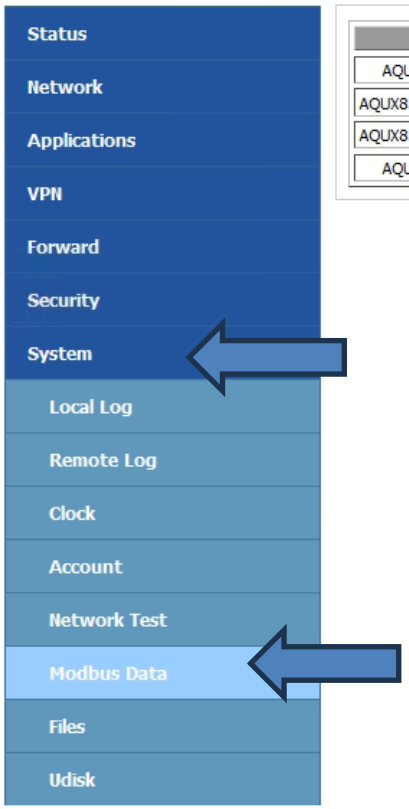
4. To save time we will load the Registers together.
   - Volts, Phase angle and Frequency for both Meters. Confirm the data modes are correct I the manual for your device. The remaining Modbus collection settings can remain the same for the testing.

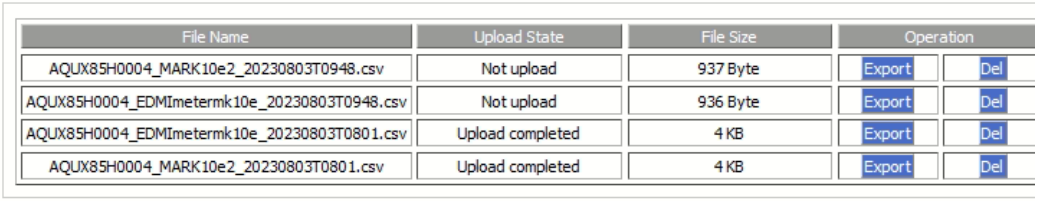






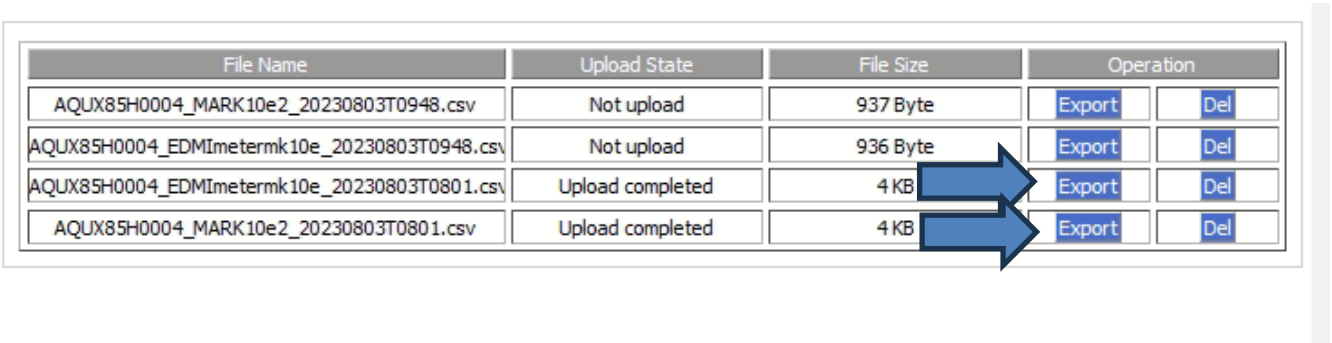
5. Give the collection process some time to scan for data. During testing the 2 .CSV files are created. One for each meter.
   - Navigate to System -> Modbus Data and confirm 2 files uploaded and 2 files collecting data. Each file has SN, name and time stamped with a date and 24-hour time stamp of when the .CSV file was created before and after uploading to FTP server.





File Name	Upload State	File Size	Operation	
AQUX85H0004_MARK10e2_20230803T0948.csv	Not upload	937 Byte	Export	Del
AQUX85H0004_EDMImetermk10e_20230803T0948.csv	Not upload	936 Byte	Export	Del
AQUX85H0004_EDMImetermk10e_20230803T0801.csv	Upload completed	4 KB	Export	Del
AQUX85H0004_MARK10e2_20230803T0801.csv	Upload completed	4 KB	Export	Del

6. Let us export the .CSV files that have been uploaded.
   - Click export on the 2 Uploaded files. We will view to confirm the data is correct.

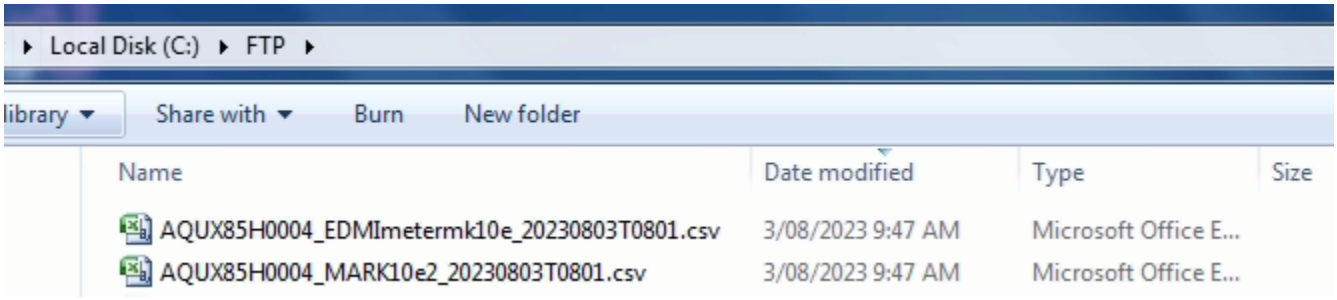


File Name	Upload State	File Size	Operation	
AQUX85H0004_MARK10e2_20230803T0948.csv	Not upload	937 Byte	Export	Del
AQUX85H0004_EDMImetermk10e_20230803T0948.csv	Not upload	936 Byte	Export	Del
AQUX85H0004_EDMImetermk10e_20230803T0801.csv	Upload completed	4 KB	Export	Del
AQUX85H0004_MARK10e2_20230803T0801.csv	Upload completed	4 KB	Export	Del

	A	B	C	D	E
1	Description	phaseCVOLTs	PHASE_C_ANGLE	FREQUENCY_Hz	
2	Address	49005	49017	49037	
3	Data Format	FLOAT32	FLOAT32	FLOAT32	
4	Data Unit	Volts	PA	Hz	
5	3/8/2023 8:01	234.875015	0	49.963001	
6	3/8/2023 8:02	235.313019	0	49.959003	
7	3/8/2023 8:03	235.557007	0	49.953003	
8	3/8/2023 8:04	235.624008	0	49.934002	
9	3/8/2023 8:05	235.307007	0	49.940002	
10	3/8/2023 8:06	235.380005	0	49.957001	
11	3/8/2023 8:08	235.435013	0	49.954002	
12	3/8/2023 8:09	235.478012	0	49.955002	
13	3/8/2023 8:10	235.362015	0	49.970001	
14	3/8/2023 8:11	235.082016	0	49.963001	
15	3/8/2023 8:12	235.447006	0	49.977001	
16	3/8/2023 8:13	235.490005	0	49.974003	

	A	B	C	D	E
1	Description	phaseCvolts	PhaseC_Angle	Frequency_HZ	
2	Address	49005	49017	49037	
3	Data Format	FLOAT32	FLOAT32	FLOAT32	
4	Data Unit	Volts	PA	Hz	
5	3/8/2023 8:00	234.965012	0	49.950001	
6	3/8/2023 8:02	235.081009	0	49.948002	
7	3/8/2023 8:03	235.343018	0	49.962002	
8	3/8/2023 8:04	234.838013	0	49.927002	
9	3/8/2023 8:05	235.361008	0	49.937004	
10	3/8/2023 8:06	235.203018	0	49.927002	
11	3/8/2023 8:08	235.19101	0	49.954002	
12	3/8/2023 8:09	235.081009	0	49.962002	
13	3/8/2023 8:10	235.063004	0	49.952003	
14	3/8/2023 8:11	235.197006	0	49.957001	
15	3/8/2023 8:12	235.19101	0	49.969002	
16	3/8/2023 8:13	234.80101	0	49.967003	
17	3/8/2023 8:15	235.063004	0	49.984001	

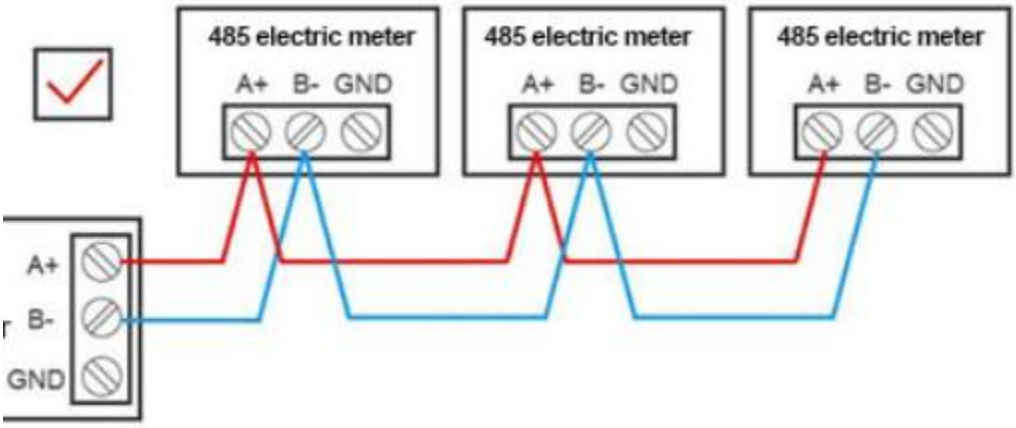
7. We can also log into the FTP server to confirm the upload is completed.



APPENDIX

RS485 Wiring with Daisy Chain connection.

Please check with manual for Meter/Device you are connecting to as the device may require a termination Resistor.



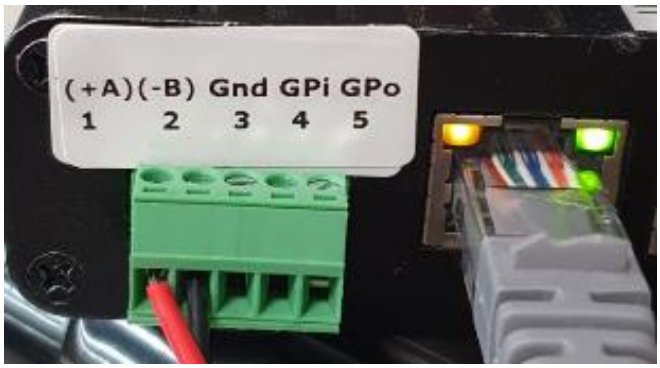
eSAM Connection 5 Pin Green port	RS-232 (DE9 or terminal block)	RS-485 (DE9 or terminal block)
TX Pin 1	TX (DE9 DTE Pin 3)	T+/A 485+ (DE9 Pin 2)
RX Pin 2	RX (DE9 DTE Pin 2)	T-/B 485- (DE9 Pin 1)
GND Pin 3	GND (DE9 Pin 5)	GND (DE9 Pin 5)
GPI Pin 4	NC	NC
GPO Pin 5	NC	NC

Note: Only the eSAM 485 & Ultra eSAM 485 variants are compatible with RS-485 signals.

RS232 Variant



RS485 Variant



ABOUT INTERCEL

Intercel is an Australian designer and manufacturer of modems and mobile equipment for use with GSM, 3G and 4G telecommunication networks. Our products provide remote connectivity for Serial and Ethernet devices in critical industrial applications. We are located in Melbourne, Australia and our head office is in Mulgrave.



ISO 9001 QMI-SAI Global

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